Appendix: Mathematical and Algorithmic Details

Appendix: Mathematical and Algorithmic Details I

*This appendix collects the formal mathematical definitions, derivations, and algorithmic specifications referenced from the main methodology section. Each subsection is self-contained; equations are labelled for cross-referencing from the body and from §**??**.*

Citation-Weighted Hypothesis Scoring Formula I

For each hypothesis H, we compute a citation-weighted evidence score aggregating all assertions relevant to H:

$$(H) = \frac{\sum_{a \in S(H)} w(a) - \sum_{a \in C(H)} w(a)}{\sum_{a \in A(H)} w(a)} \tag{1}$$

where $S(H)$ is the set of supporting assertions, $C(H)$ the set of contradicting assertions, $A(H)$ all assertions for H (including neutral), and the weight function is

$$w(a) = \log(1 + \text{citations}(a)) \cdot \text{confidence}(a). \tag{2}$$

Citation-Weighted Hypothesis Scoring Formula II

The logarithmic citation weighting assigns higher weight to papers that have accumulated more citations within the retrieved corpus window. This measures **published attention and visibility** in the literature graph—not truth, popularity on social media, field size alone, or citation-network centrality (PageRank is not used). We report sensitivity analyses under uniform weighting, confidence-only, raw citation (popularity stress test), age discount, and field-normalized cohort weighting; rank-stability Spearman $\rho = 0.762$ versus the default policy. The score lies in $[-1, 1]$: values indicate citation-weighted evidence mapping and hypothesis triage within the corpus, not calibrated confirmation of scientific truth.

Temporal aggregation. We additionally compute temporal trends by evaluating the cumulative score at each year t, using only assertions from papers published in year $\leq t$:

Citation-Weighted Hypothesis Scoring Formula III

$$(H, t) = \frac{\sum_{a \in S(H,t)} w(a) - \sum_{a \in C(H,t)} w(a)}{\sum_{a \in A(H,t)} w(a)}. \tag{3}$$

This reveals whether support for a hypothesis is growing, declining, or plateauing over time. Cumulative aggregation (rather than yearly windows) is preferred because per-year assertion counts for narrow hypotheses are too sparse for stable point estimates.

Algorithmic specification. The scoring routine is a pure function of the assertion set; it has no hidden state and is deterministic given the input. The reference implementation lives in `src/knowledge_graph/hypothesis.py` with weight policies in `src/knowledge_graph/hypothesis_weights.py`:

Citation-Weighted Hypothesis Scoring Formula IV

```
function score(H, assertions):
    S, C, A_all = 0, 0, 0
    for a in assertions where a.hypothesis == H:
        w = log(1 + a.citations) * a.confidence
        if a.direction == "supports":     S     += w
        elif a.direction == "contradicts": C     += w
        A_all += w
    return (S - C) / A_all  if A_all > 0  else 0
```

Boundary tests in `tests/test_scoring.py` confirm that all-support fixtures yield $+1$, all-contradict fixtures yield -1, and balanced fixtures yield 0 within numerical tolerance.

Non-negative Matrix Factorization (NMF) for Topic Modeling I

We apply NMF to the TF-IDF matrix of the corpus to discover latent topics. Given the document-term matrix $V \in \mathbb{R}_{\geq 0}^{n \times m}$, NMF finds factor matrices $W \in \mathbb{R}_{\geq 0}^{n \times k}$ and $H \in \mathbb{R}_{\geq 0}^{k \times m}$ such that $V \approx WH$, where k is the number of topics. We use multiplicative update rules [lee1999nmf]:

$$H \leftarrow H \odot \frac{W^T V}{W^T W H + \epsilon}, \qquad W \leftarrow W \odot \frac{V H^T}{W H H^T + \epsilon}, \tag{4}$$

with $\epsilon = 10^{-10}$ for numerical stability and a fixed random seed of 42 for reproducibility (deterministic topic alignment across pipeline runs, with empirical stability confirmed via Jaccard similarities > 0.90 across alternative seeds).

Non-negative Matrix Factorization (NMF) for Topic Modeling II

Term-Frequency Inverse Document Frequency (TF-IDF). The document-term matrix is constructed using a smoothed TF-IDF weighting [salton1975vector]. For term t in document d:

$$\text{TF-IDF}(t,d) = \text{tf}(t,d) \cdot \left[\log\left(\frac{N}{\text{df}(t)+1}\right) + 1\right], \tag{5}$$

where $\text{tf}(t,d) = \text{count}(t,d)/|d|$ is the normalized term frequency, N the total number of documents, and $\text{df}(t)$ the document frequency of term t. The $+1$ additive smoothing in the denominator prevents division by zero and reduces the weight of extremely rare terms; the outer $+1$ ensures strictly positive IDF values. Document vectors are L2-normalized before NMF factorization.

Field Growth-Rate Estimation I

The **mean year-over-year growth rate** $\bar{g}$ is the arithmetic mean of annual growth rates computed only for years where the prior year had non-zero publications:

$$\bar{g} = \frac{1}{|Y|} \sum_{y \in Y} \frac{n_y - n_{y-1}}{n_{y-1}}, \tag{6}$$

where $Y = \{y : n_{y-1} > 0\}$ and n_y is the number of publications in year y.

The **doubling time** t_d is derived from the mean annual growth rate:

$$t_d = \frac{\ln 2}{\ln(1 + \bar{g})}. \tag{7}$$

Field Growth-Rate Estimation II

The **compound annual growth rate** (CAGR) over the full span $[y_0, y_T]$ is

$$= \left(\frac{n_{\text{cumulative}}(y_T)}{n_{\text{cumulative}}(y_0)} \right)^{1/(y_T - y_0)} - 1. \tag{8}$$

For the current corpus, $= 20.36\%$. The more recent growth phase (2010–2026) exhibits substantially higher annualized growth than the long-run average; reporting both the T-year CAGR and recent-phase CAGR avoids overstating maturity-era expansion.

Advanced Visualization Methods I

PCA of TF-IDF Embeddings

Principal Component Analysis (PCA) is applied to the TF-IDF matrix V to project each document into a 2-D space. The projection preserves the directions of maximum variance, enabling visual inspection of document clustering by domain. Loading arrows overlay the top-variance terms onto the scatter plot, showing which vocabulary drives the principal components.

Hierarchical Clustering Dendrogram

For each domain s, we compute the centroid $\bar{v}_s = \frac{1}{|D_s|} \sum_{d \in D_s} v_d$ where D_s is the set of documents in domain s and v_d is the TF-IDF vector of document d. Ward linkage is applied to the centroid matrix to produce a hierarchical clustering dendrogram showing semantic proximity between domains.

Advanced Visualization Methods II

Term Heatmap

For each domain s and term t, we compute the mean TF-IDF weight $\bar{w}_{s,t} = \frac{1}{|D_s|} \sum_{d \in D_s} \text{TF-IDF}(t, d)$. The heatmap displays $\bar{w}_{s,t}$ for the top-k terms (by global document frequency) across all domains, with cell intensity proportional to mean weight. This reveals distinctive vocabulary patterns that differentiate domains beyond the keyword-level classification used for subfield assignment.

Term Co-occurrence Matrix

The co-occurrence matrix $C \in \mathbb{R}^{k \times k}$ counts the number of documents in which two terms appear together. For top-k terms by document frequency, $C_{ij} = |\{d : t_i \in d \wedge t_j \in d\}|$. The matrix is normalized to $[0, 1]$ by dividing by the maximum entry and visualized as a symmetric heatmap.

Nanopublication RDF Schema I

Each nanopublication is serialized to RDF/TriG per the nanopublication standard [groth2010anatomy, kuhn2016decentralized], producing four named graphs. The following annotated example illustrates the structure for a single assertion:

```
@prefix np:   <http://www.nanopub.org/nschema#> .
@prefix prov: <http://www.w3.org/ns/prov#> .
@prefix dc:   <http://purl.org/dc/terms/> .
@prefix aif:  <http://activeinference.institute/ontology/>
@prefix xsd:  <http://www.w3.org/2001/XMLSchema#> .

# HEAD GRAPH: links nanopub to its three component graphs
<http://activeinference.institute/nanopub/a1b2c3d4e5f6#head
  <http://activeinference.institute/nanopub/a1b2c3d4e5f6>
    a np:Nanopublication ;
    np:hasAssertion       <...#assertion> ;
```

Nanopublication RDF Schema II

```
    np:hasProvenance      <...#provenance> ;
    np:hasPublicationInfo <...#pubinfo> .
}

# ASSERTION GRAPH: the core scientific claim
<http://activeinference.institute/nanopub/a1b2c3d4e5f6#asse
  aif:paper/10.1038_nrn2787 aif:asserts aif:assertion/a1b2c
  aif:assertion/a1b2c3
    aif:supports       aif:hypothesis/fep_universality ;
    aif:claim          "The paper provides foundational sup
                        unified brain theory."^^xsd:string
    aif:confidence     "0.85"^^xsd:double ;
    aif:citationCount  "5000"^^xsd:integer .
}

# PROVENANCE GRAPH: extraction lineage
<http://activeinference.institute/nanopub/a1b2c3d4e5f6#prov
```

Nanopublication RDF Schema III

```
  aif:assertion/a1b2c3
    prov:wasGeneratedBy   <http://activeinference.institute
    prov:generatedAtTime  "2026-01-15T12:00:00+00:00"^^xsd:
    prov:wasAttributedTo  "act_inf_metaanalysis/gemma3:4b"^
    prov:hadPrimarySource aif:paper/10.1038_nrn2787 .
}

# PUBLICATION INFO GRAPH: nanopublication metadata
<http://activeinference.institute/nanopub/a1b2c3d4e5f6#pub
  <http://activeinference.institute/nanopub/a1b2c3d4e5f6>
    dc:created "2026-01-15T12:00:00+00:00"^^xsd:dateTime ;
    dc:creator "act_inf_metaanalysis/gemma3:4b"^^xsd:string
    dc:license <https://creativecommons.org/publicdomain/ze
}
```

Nanopublication RDF Schema IV

Namespace Definitions

Table 1: RDF namespace definitions used in the knowledge graph and nanopublication serialization. Each prefix maps to a W3C or domain-specific URI.

Prefix	URI	Purpose
`np:`	`http://www.nanopub.org/nschema#`	Nanopub structural predicates
`prov:`	`http://www.w3.org/ns/prov#`	PROV-O provenance model
`dc:`	`http://purl.org/dc/terms/`	Dublin Core metadata
`aif:`	`http://activeinference.institute/ontology/`	Domain-specific predicates
`xsd:`	`http://www.w3.org/2001/XMLSchema#`	XML Schema datatypes

Core Triple Patterns

The knowledge graph encodes five fundamental relationships:

Table 2: Core RDF triple patterns encoding the five fundamental relationships in the knowledge graph. Each pattern links paper, assertion, hypothesis, or subfield nodes.

Triple Pattern	Meaning
`Paper--aif:asserts-->Assertion`	A paper makes a claim
`Paper--aif:cites-->Paper`	Intra-corpus citation link
`Paper--aif:belongsTo-->Subfield`	Domain classification
`Assertion--aif:supports-->Hypothesis`	Supporting evidence
`Assertion--aif:contradicts-->Hypothesis`	Contradicting evidence